

Science | Volume 393 | Issue 6817

Issue date: 2026-09-17 | Retrieved: 2026-09-17T22:53:06.370676+00:00 | 11 relevant primary-research papers

Evidence discipline

Paper-supported findings, published enabling evidence, cross-species inference, speculative hypotheses, and proposed pilot experiments are labeled separately. Access notes state when only an abstract or limited preview was available.

Screening notes

- Official publisher issue page returned HTTP 403; no full-page review is claimed.
- Screened 11 biology-relevant journal articles after excluding reviews, perspectives, news, corrections, and editorials by metadata/title rules.
- All summaries are deterministic extractions from the bounded public evidence packet; no language-model API or paid research service was used.

Official issue

<https://www.science.org/toc/science/393/6817>

Why Xenopus now

Early Xenopus combines rapid external development, targeted microinjection, explants, lineage tracing, live imaging, F0 CRISPR, rescue, and scalable replication. It can convert abstract-supported candidate mechanisms into spatially resolved causal tests before committing to slower mammalian models.

Transcriptome-based classification in mice with ASD-risk mutations

Junyeop Daniel Roh, Yukyung Jun, Heesu Jeon, Junyoung Kim, Yunho Yi, Minji Kim, Heejin Cho, Yusang Oh, et al.

journal-article | Science 393(6817) | 2026-09-17

DOI 10.1126/science.adz6688

Access: Publisher-deposited abstract retrieved through Crossref; no full-text review is claimed. | Evidence status: abstract-supported

MAIN DISCOVERY	We report a sex-balanced atlas of 1008 prefrontal RNA sequencing (RNA-seq) profiles from 17 mouse lines carrying ASD-risk mutations.
WHY IT MATTERS	The framework classifies independent mouse lines and identifies subgroups with conserved synaptic directionality, supporting molecular stratification.
MAIN / NEW METHODS	• Single-nucleus multi-omics • Computational modeling
KEY EVIDENCE	• We report a sex-balanced atlas of 1008 prefrontal RNA sequencing (RNA-seq) profiles from 17 mouse lines carrying ASD-risk mutations. • Single-nucleus RNA-seq revealed broader cell type remodeling in group 1 than in group 2, and cell type-specific modules showed reciprocal associations that mirrored bulk transcriptomic signatures. • The framework classifies independent mouse lines and identifies subgroups with conserved synaptic directionality, supporting molecular stratification.
LIMITATIONS	• This automated report used title, DOI metadata, and an available abstract; it does not claim manual full-text, figure, or complete methods review. • The deterministic extractor cannot assess experimental quality, effect size, statistics, or unreported negative results beyond the retrieved source text. • The proposed Xenopus mechanism is new speculation and has not been established by the featured paper.

Developmental timing and tissue autonomy of Transcriptome-based classification in mice with ASD-risk mutations

IMPORTANCE / SIGNIFICANCE

Gene Regulation And Chromatin is central to how embryonic cells convert molecular state into reproducible tissue organization. Establishing whether the featured mechanism operates during early vertebrate development could reveal a conserved control point and a tractable route to explain developmental variability without overstating clinical translation.

KNOWLEDGE GAP

The featured paper does not establish whether the focal regulator or mechanism described by the featured paper acts during early vertebrate development, whether its effect is cell-autonomous, which developmental window is sensitive, or whether molecular and morphogenetic phenotypes can be rescued independently.

RATIONALE AND XENOPUS ADVANTAGE

The paper supplies a candidate mechanism, while early *Xenopus* provides rapid external development, targeted blastomere injection, explants, lineage tracing, live imaging, inexpensive biological replication across clutches, and direct comparison of molecular and tissue-scale outcomes.

CENTRAL HYPOTHESIS

Speculative: the focal regulator or mechanism described by the featured paper acts cell-autonomously during stages 8-14 to stabilize pathway activity, lineage allocation, and spatial marker-domain size; acute loss will reduce the predicted molecular readout and increase pattern variance, while stage-matched wild-type rescue will restore both.

EXPERIMENTAL DESIGN

Use *Xenopus tropicalis* for F0 genetics and *Xenopus laevis* explants. Target animal-cap, organizer, and mesendodermal lineages by injecting two independent CRISPR RNPs at the 4- to 8-cell stage with a fluorescent lineage tracer; analyze stages 8-14. Groups: uninjected, non-targeting RNP, each guide alone, pooled guides, dose series, wild-type mRNA rescue, perturbation-resistant rescue, and temporally delayed perturbation where a validated small molecule or inducible construct exists. Primary readouts: pathway activity, lineage allocation, and spatial marker-domain size. Secondary readouts: cell behavior, developmental timing, and gastrulation morphology, amplicon sequencing, RT-qPCR or in situ hybridization, and an orthogonal biochemical or reporter assay. Use at least three independent spawnings, 20-30 embryos per group per spawning for morphology, and 8-10 pooled or individually profiled samples per molecular condition; randomize embryos, blind image scoring, and analyze clutch as a random effect with multiplicity-controlled planned contrasts. Decision point: advance to epistasis and spatial single-cell profiling only if both guides reproduce the effect and rescue restores at least 50% of the molecular and developmental phenotype.

EXPECTED RESULTS AND INTERPRETATIONS

Support would be a guide-concordant loss of the predicted molecular readout, increased spatial or embryo-to-embryo variance, and rescue of both molecular and tissue phenotypes. This would place the mechanism upstream of developmental robustness. A molecular phenotype without morphogenesis would suggest buffering or a parallel tissue effector and would redirect the next aim to sensitized or explant conditions. A morphogenetic phenotype without the predicted reporter change would argue for an alternative biochemical function. No reproducible phenotype after confirmed editing would refute an obligate early role and shift the model toward redundancy, later function, or species-specific deployment.

POTENTIAL PITFALLS

- F0 mosaicism may weaken or spatially blur phenotypes.
- Paralog redundancy or maternal stores may mask an early requirement.

- High-dose perturbation may cause nonspecific toxicity or developmental delay.
- The featured mechanism may not be conserved or may use different effectors in amphibians.
- Abstract-only evidence may omit critical context needed to choose the most specific assay.

ALTERNATIVE STRATEGIES

- Map editing per embryo and enrich analysis for high-editing specimens; establish stable alleles if mosaicism remains limiting.
- Target expressed paralogs jointly, use maternal-zygotic strategies where feasible, or test dominant-negative and degron-based acute perturbations.
- Use lower doses, later induction, tissue-restricted injection, explants, and matched survival/staging controls to separate mechanism from toxicity.
- Test *Xenopus laevis* and *X. tropicalis* orthologs, then validate the key interaction biochemically or in a complementary cell model.
- Before full-scale work, manually review the linked paper's figures and methods and revise the reporter or perturbation to match the demonstrated mechanism.

PUBLISHED ENABLING EVIDENCE AND PROPOSED PRELIMINARY-DATA PLAN

Published enabling evidence and proposed preliminary-data plan: no unpublished user-laboratory data are assumed. The featured paper is the cited cross-species enabling source; direct *Xenopus* evidence was not automatically verified. Pilot 1: map candidate expression or pathway activity at 5-6 early stages using RT-qPCR and in situ hybridization (three spawnings; expected Figure 1A-B); go if a reproducible spatial or temporal signal exceeds twofold over the lowest stage. Pilot 2: test two CRISPR RNPs plus non-targeting control in three spawnings with at least 25 embryos per group, amplicon sequencing, blinded morphology, and the primary molecular readout (Figure 1C-E); go if both guides produce the same directional effect of at least 20% with acceptable survival. Pilot 3: perform wild-type mRNA rescue in three spawnings with at least 20 embryos per group (Figure 1F-G); go if rescue reverses at least 50% of the molecular and phenotype effect. Failure at any threshold triggers the paired alternative strategy rather than a claim of preliminary support.

NOVELTY medium-high - transfers a recent mechanism into a spatially and temporally tractable vertebrate embryo	FEASIBILITY high for the pilots - injection, explants, imaging, genotyping, and rescue are routine <i>Xenopus</i> capabilities	RISK medium - conservation, maternal contribution, and mosaicism are explicit decision points
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GRANT FIT

Exploratory Aim 1 for conservation and causality; successful pilots support Aim 2 epistasis and Aim 3 spatial mechanism.

STRONGEST REVIEWER-FACING PREMISE

Xenopus can test when, where, and whether a newly reported cellular mechanism controls intact vertebrate development in one experimentally integrated system.

SUPPORTING SOURCES

featured paper: [Featured paper \(publisher-linked source\) 10.1126/science.adz6688](https://doi.org/10.1126/science.adz6688)

Splenic regulation of systemic platelet activation state

Lisa Laun, Alexander Leunig, Felix Zhang, Sezer Akgöl, Dario Rossaro, Matthias P. Fabritius, Afra Anjum, Lennart Kreutz, et al.

journal-article | Science 393(6817) | 2026-09-17

DOI 10.1126/science.aec7230

Access: Publisher-deposited abstract retrieved through Crossref; no full-text review is claimed. | Evidence status: abstract-supported

MAIN DISCOVERY	In mice, we found that the spleen filters activated platelets in thrombotic and inflammatory conditions, limiting pathology by maintaining platelet quiescence.
WHY IT MATTERS	The study advances mechanistic understanding of Splenic regulation of systemic platelet activation state and creates a defined basis for testing conservation and developmental function in a vertebrate embryo.
MAIN / NEW METHODS	• Publisher-deposited title/abstract analysis; consult the linked paper for complete technical details
KEY EVIDENCE	• In mice, we found that the spleen filters activated platelets in thrombotic and inflammatory conditions, limiting pathology by maintaining platelet quiescence.
LIMITATIONS	• This automated report used title, DOI metadata, and an available abstract; it does not claim manual full-text, figure, or complete methods review. • The deterministic extractor cannot assess experimental quality, effect size, statistics, or unreported negative results beyond the retrieved source text. • The proposed Xenopus mechanism is new speculation and has not been established by the featured paper.

Developmental timing and tissue autonomy of Splenic regulation of systemic platelet activation state

IMPORTANCE / SIGNIFICANCE

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KNOWLEDGE GAP

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CENTRAL HYPOTHESIS

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EXPERIMENTAL DESIGN

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ALTERNATIVE STRATEGIES

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NOVELTY medium-high - transfers a recent mechanism into a spatially and temporally tractable vertebrate embryo	FEASIBILITY high for the pilots - injection, explants, imaging, genotyping, and rescue are routine <i>Xenopus</i> capabilities	RISK medium - conservation, maternal contribution, and mosaicism are explicit decision points
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GRANT FIT

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STRONGEST REVIEWER-FACING PREMISE

Xenopus can test when, where, and whether a newly reported cellular mechanism controls intact vertebrate development in one experimentally integrated system.

SUPPORTING SOURCES

featured paper: [Featured paper \(publisher-linked source\) 10.1126/science.aec7230](https://doi.org/10.1126/science.aec7230)

A noncontiguous code for RNA-guided DNA recognition at the origin of CRISPR-Cas

Peter H. Yoon, Kenneth J. Loi, Zeyuan Terry Zhang, Trevor A. Docter, Santiago C. Lopez, Conner J. Langeberg, Muhammad Moez Ur-Rehman, Kamakshi Vohra, et al.

journal-article | Science 393(6817) | 2026-09-17

DOI 10.1126/science.aei0498

Access: Publisher-deposited abstract retrieved through Crossref; no full-text review is claimed. | Evidence status: abstract-supported

MAIN DISCOVERY	In this study, we report the discovery of Viral Interference Programmable Repeat (VIPR) systems consisting of a Vipr protein ancestral to the earliest CRISPR-Cas effectors and VIPR RNAs (vrRNAs) comprising alternating GGY/NN motifs.
WHY IT MATTERS	These results suggest that adaptive immunity originated from ancient warfare between viruses, revealing a previously unidentified logic for encoding information in sequence.
MAIN / NEW METHODS	• CRISPR genome editing
KEY EVIDENCE	• CRISPR-Cas provides RNA-mediated adaptive immunity, but how its first RNA-guided effector arose is unclear. • In this study, we report the discovery of Viral Interference Programmable Repeat (VIPR) systems consisting of a Vipr protein ancestral to the earliest CRISPR-Cas effectors and VIPR RNAs (vrRNAs) comprising alternating GGY/NN motifs. • Natural vrRNA targets suggest that VIPR systems act against competing phages, and we demonstrate programmable phage defense by redirecting the complex for transcriptional repression.
LIMITATIONS	• This automated report used title, DOI metadata, and an available abstract; it does not claim manual full-text, figure, or complete methods review. • The deterministic extractor cannot assess experimental quality, effect size, statistics, or unreported negative results beyond the retrieved source text. • The proposed Xenopus mechanism is new speculation and has not been established by the featured paper.

Developmental timing and tissue autonomy of A noncontiguous code for RNA-guided DNA recognition at the origin of CRISPR-Cas

IMPORTANCE / SIGNIFICANCE

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KNOWLEDGE GAP

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RATIONALE AND XENOPUS ADVANTAGE

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CENTRAL HYPOTHESIS

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EXPERIMENTAL DESIGN

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POTENTIAL PITFALLS

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ALTERNATIVE STRATEGIES

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NOVELTY medium-high - transfers a recent mechanism into a spatially and temporally tractable vertebrate embryo	FEASIBILITY high for the pilots - injection, explants, imaging, genotyping, and rescue are routine <i>Xenopus</i> capabilities	RISK medium - conservation, maternal contribution, and mosaicism are explicit decision points
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GRANT FIT

Exploratory Aim 1 for conservation and causality; successful pilots support Aim 2 epistasis and Aim 3 spatial mechanism.

STRONGEST REVIEWER-FACING PREMISE

Xenopus can test when, where, and whether a newly reported cellular mechanism controls intact vertebrate development in one experimentally integrated system.

SUPPORTING SOURCES

featured paper: [Featured paper \(publisher-linked source\) 10.1126/science.aei0498](https://doi.org/10.1126/science.aei0498)

A viral origin for RNA-guided immunity

Jack P. K. Bravo

journal-article | Science 393(6817) | 2026-09-17

DOI 10.1126/science.ael0758

Access: Publisher-deposited abstract retrieved through Crossref; no full-text review is claimed. | Evidence status: abstract-supported

MAIN DISCOVERY	Ancient viral warfare could be at the root of modern class 1 CRISPR bacterial defense systems
WHY IT MATTERS	The study advances mechanistic understanding of A viral origin for RNA-guided immunity and creates a defined basis for testing conservation and developmental function in a vertebrate embryo.
MAIN / NEW METHODS	• CRISPR genome editing
KEY EVIDENCE	• The retrieved publisher-deposited metadata identifies this item as a journal article in the detected volume and issue. • The DOI provides the canonical path for manual verification of figures and full methods.
LIMITATIONS	• This automated report used title, DOI metadata, and an available abstract; it does not claim manual full-text, figure, or complete methods review. • The deterministic extractor cannot assess experimental quality, effect size, statistics, or unreported negative results beyond the retrieved source text. • The proposed Xenopus mechanism is new speculation and has not been established by the featured paper.

Developmental timing and tissue autonomy of A viral origin for RNA-guided immunity

IMPORTANCE / SIGNIFICANCE

Immunity And Host Response is central to how embryonic cells convert molecular state into reproducible tissue organization. Establishing whether the featured mechanism operates during early vertebrate development could reveal a conserved control point and a tractable route to explain developmental variability without overstating clinical translation.

KNOWLEDGE GAP

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CENTRAL HYPOTHESIS

Speculative: the focal regulator or mechanism described by the featured paper acts cell-autonomously during stages 8-18 to stabilize cell-stress signaling, epithelial integrity, and lineage-specific transcription; acute loss will reduce the predicted molecular readout and increase pattern variance, while stage-matched wild-type rescue will restore both.

EXPERIMENTAL DESIGN

Use *Xenopus tropicalis* for F0 genetics and *Xenopus laevis* explants. Target epidermal and mesendodermal lineages before mature immunity by injecting two independent CRISPR RNPs at the 4- to 8-cell stage with a fluorescent lineage tracer; analyze stages 8-18. Groups: uninjected, non-targeting RNP, each guide alone, pooled guides, dose series, wild-type mRNA rescue, perturbation-resistant rescue, and temporally delayed perturbation where a validated small molecule or inducible construct exists. Primary readouts: cell-stress signaling, epithelial integrity, and lineage-specific transcription. Secondary readouts: survival, barrier morphology, and developmental timing, amplicon sequencing, RT-qPCR or in situ hybridization, and an orthogonal biochemical or reporter assay. Use at least three independent spawnings, 20-30 embryos per group per spawning for morphology, and 8-10 pooled or individually profiled samples per molecular condition; randomize embryos, blind image scoring, and analyze clutch as a random effect with multiplicity-controlled planned contrasts. Decision point: advance to epistasis and spatial single-cell profiling only if both guides reproduce the effect and rescue restores at least 50% of the molecular and developmental phenotype.

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Ancient DNA unveils distinctive ancestries in the Bronze and Iron Ages of East Tianshan

Chao Ning, Ke Wang, Tianyi Wang, Jian Ma, Xue Zhao, Tongyuan Xi, Yongqiang Wang, Peng Cao, et al.

journal-article | Science 393(6817) | 2026-09-17

DOI 10.1126/science.adv1295

Access: Publisher-deposited abstract retrieved through Crossref; no full-text review is claimed. | Evidence status: abstract-supported

MAIN DISCOVERY	Here we report genome-wide data from 135 ancient individuals from 11 archaeological sites.
WHY IT MATTERS	The study advances mechanistic understanding of Ancient DNA unveils distinctive ancestries in the Bronze and Iron Ages of East Tianshan and creates a defined basis for testing conservation and developmental function in a vertebrate embryo.
MAIN / NEW METHODS	• Publisher-deposited title/abstract analysis; consult the linked paper for complete technical details
KEY EVIDENCE	• Here we report genome-wide data from 135 ancient individuals from 11 archaeological sites. • We identify a previously unrecognized Bronze Age admixture between populations related to Yellow River millet farmers and steppe pastoralists associated with the Chemurchek culture. • In contrast, we find little genetic contribution from contemporaneous middle-to-late Bronze Age steppe pastoralists, despite their eastward expansion across the Eurasian Steppe.
LIMITATIONS	• This automated report used title, DOI metadata, and an available abstract; it does not claim manual full-text, figure, or complete methods review. • The deterministic extractor cannot assess experimental quality, effect size, statistics, or unreported negative results beyond the retrieved source text. • The proposed Xenopus mechanism is new speculation and has not been established by the featured paper.

Developmental timing and tissue autonomy of Ancient DNA unveils distinctive ancestries in the Bronze and Iron Ages of East...

IMPORTANCE / SIGNIFICANCE

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EXPERIMENTAL DESIGN

Use *Xenopus tropicalis* for F0 genetics and *Xenopus laevis* explants. Target animal-cap, organizer, and mesendodermal lineages by injecting two independent CRISPR RNPs at the 4- to 8-cell stage with a fluorescent lineage tracer; analyze stages 8-14. Groups: uninjected, non-targeting RNP, each guide alone, pooled guides, dose series, wild-type mRNA rescue, perturbation-resistant rescue, and temporally delayed perturbation where a validated small molecule or inducible construct exists. Primary readouts: pathway activity, lineage allocation, and spatial marker-domain size. Secondary readouts: cell behavior, developmental timing, and gastrulation morphology, amplicon sequencing, RT-qPCR or in situ hybridization, and an orthogonal biochemical or reporter assay. Use at least three independent spawnings, 20-30 embryos per group per spawning for morphology, and 8-10 pooled or individually profiled samples per molecular condition; randomize embryos, blind image scoring, and analyze clutch as a random effect with multiplicity-controlled planned contrasts. Decision point: advance to epistasis and spatial single-cell profiling only if both guides reproduce the effect and rescue restores at least 50% of the molecular and developmental phenotype.

EXPECTED RESULTS AND INTERPRETATIONS

Support would be a guide-concordant loss of the predicted molecular readout, increased spatial or embryo-to-embryo variance, and rescue of both molecular and tissue phenotypes. This would place the mechanism upstream of developmental robustness. A molecular phenotype without morphogenesis would suggest buffering or a parallel tissue effector and would redirect the next aim to sensitized or explant conditions. A morphogenetic phenotype without the predicted reporter change would argue for an alternative biochemical function. No reproducible phenotype after confirmed editing would refute an obligate early role and shift the model toward redundancy, later function, or species-specific deployment.

POTENTIAL PITFALLS

- F0 mosaicism may weaken or spatially blur phenotypes.
- Paralog redundancy or maternal stores may mask an early requirement.

- High-dose perturbation may cause nonspecific toxicity or developmental delay.
- The featured mechanism may not be conserved or may use different effectors in amphibians.
- Abstract-only evidence may omit critical context needed to choose the most specific assay.

ALTERNATIVE STRATEGIES

- Map editing per embryo and enrich analysis for high-editing specimens; establish stable alleles if mosaicism remains limiting.
- Target expressed paralogs jointly, use maternal-zygotic strategies where feasible, or test dominant-negative and degron-based acute perturbations.
- Use lower doses, later induction, tissue-restricted injection, explants, and matched survival/staging controls to separate mechanism from toxicity.
- Test *Xenopus laevis* and *X. tropicalis* orthologs, then validate the key interaction biochemically or in a complementary cell model.
- Before full-scale work, manually review the linked paper's figures and methods and revise the reporter or perturbation to match the demonstrated mechanism.

PUBLISHED ENABLING EVIDENCE AND PROPOSED PRELIMINARY-DATA PLAN

Published enabling evidence and proposed preliminary-data plan: no unpublished user-laboratory data are assumed. The featured paper is the cited cross-species enabling source; direct *Xenopus* evidence was not automatically verified. Pilot 1: map candidate expression or pathway activity at 5-6 early stages using RT-qPCR and in situ hybridization (three spawnings; expected Figure 1A-B); go if a reproducible spatial or temporal signal exceeds twofold over the lowest stage. Pilot 2: test two CRISPR RNPs plus non-targeting control in three spawnings with at least 25 embryos per group, amplicon sequencing, blinded morphology, and the primary molecular readout (Figure 1C-E); go if both guides produce the same directional effect of at least 20% with acceptable survival. Pilot 3: perform wild-type mRNA rescue in three spawnings with at least 20 embryos per group (Figure 1F-G); go if rescue reverses at least 50% of the molecular and phenotype effect. Failure at any threshold triggers the paired alternative strategy rather than a claim of preliminary support.

NOVELTY medium-high - transfers a recent mechanism into a spatially and temporally tractable vertebrate embryo	FEASIBILITY high for the pilots - injection, explants, imaging, genotyping, and rescue are routine <i>Xenopus</i> capabilities	RISK medium - conservation, maternal contribution, and mosaicism are explicit decision points
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GRANT FIT

Exploratory Aim 1 for conservation and causality; successful pilots support Aim 2 epistasis and Aim 3 spatial mechanism.

STRONGEST REVIEWER-FACING PREMISE

Xenopus can test when, where, and whether a newly reported cellular mechanism controls intact vertebrate development in one experimentally integrated system.

SUPPORTING SOURCES

featured paper: [Featured paper \(publisher-linked source\) 10.1126/science.adv1295](https://doi.org/10.1126/science.adv1295)

Parsing heterogeneity in autism spectrum disorder

Jacqueline M. Martín, Daniel H. Geschwind

journal-article | Science 393(6817) | 2026-09-17

DOI 10.1126/science.ael3062

Access: Publisher-deposited abstract retrieved through Crossref; no full-text review is claimed. | Evidence status: abstract-supported

MAIN DISCOVERY	Stratifying genetic mutations points to different pathophysiological pathways to a neurodevelopmental disorder
WHY IT MATTERS	The study advances mechanistic understanding of Parsing heterogeneity in autism spectrum disorder and creates a defined basis for testing conservation and developmental function in a vertebrate embryo.
MAIN / NEW METHODS	• Publisher-deposited title/abstract analysis; consult the linked paper for complete technical details
KEY EVIDENCE	• The retrieved publisher-deposited metadata identifies this item as a journal article in the detected volume and issue. • The DOI provides the canonical path for manual verification of figures and full methods.
LIMITATIONS	• This automated report used title, DOI metadata, and an available abstract; it does not claim manual full-text, figure, or complete methods review. • The deterministic extractor cannot assess experimental quality, effect size, statistics, or unreported negative results beyond the retrieved source text. • The proposed Xenopus mechanism is new speculation and has not been established by the featured paper.

Developmental timing and tissue autonomy of Parsing heterogeneity in autism spectrum disorder

IMPORTANCE / SIGNIFICANCE

Gene Regulation And Chromatin is central to how embryonic cells convert molecular state into reproducible tissue organization. Establishing whether the featured mechanism operates during early vertebrate development could reveal a conserved control point and a tractable route to explain developmental variability without overstating clinical translation.

KNOWLEDGE GAP

The featured paper does not establish whether the focal regulator or mechanism described by the featured paper acts during early vertebrate development, whether its effect is cell-autonomous, which developmental window is sensitive, or whether molecular and morphogenetic phenotypes can be rescued independently.

RATIONALE AND XENOPUS ADVANTAGE

The paper supplies a candidate mechanism, while early *Xenopus* provides rapid external development, targeted blastomere injection, explants, lineage tracing, live imaging, inexpensive biological replication across clutches, and direct comparison of molecular and tissue-scale outcomes.

CENTRAL HYPOTHESIS

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EXPERIMENTAL DESIGN

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GRANT FIT

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SUPPORTING SOURCES

featured paper: [Featured paper \(publisher-linked source\) 10.1126/science.ael3062](https://doi.org/10.1126/science.ael3062)

Germline *EGFR* T790M mutation and lung cancer risk

Jaclyn LoPiccolo, Steven Micheletti, Jing Shi, Wei Wang, Shubham Saini, Keng-Han Lin, Wanwan Xu, Pierre Fontanillas, et al.

journal-article | Science 393(6817) | 2026-09-17

DOI 10.1126/science.aec0473

Access: Publisher-deposited abstract retrieved through Crossref; no full-text review is claimed. | Evidence status: abstract-supported

MAIN DISCOVERY	Global geographic and ancestry analyses show higher T790M prevalence in the US than in British- and Irish-descendant populations, reflecting a Southern Appalachian founder event about 200 to 225 years ago, increasing regional prevalence and affecting those of British, Irish, and African descent.
WHY IT MATTERS	The study advances mechanistic understanding of Germline <i>EGFR</i> T790M mutation and lung cancer risk and creates a defined basis for testing conservation and developmental function in a vertebrate embryo.
MAIN / NEW METHODS	Genome-wide association or QTL mapping
KEY EVIDENCE	Global geographic and ancestry analyses show higher T790M prevalence in the US than in British- and Irish-descendant populations, reflecting a Southern Appalachian founder event about 200 to 225 years ago, increasing regional prevalence and affecting those of British, Irish, and African descent.
LIMITATIONS	- This automated report used title, DOI metadata, and an available abstract; it does not claim manual full-text, figure, or complete methods review. - The deterministic extractor cannot assess experimental quality, effect size, statistics, or unreported negative results beyond the retrieved source text. - The proposed Xenopus mechanism is new speculation and has not been established by the featured paper.

Developmental timing and tissue autonomy of Germline *EGFR* T790M mutation and lung cancer risk

IMPORTANCE / SIGNIFICANCE

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KNOWLEDGE GAP

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RATIONALE AND XENOPUS ADVANTAGE

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CENTRAL HYPOTHESIS

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EXPERIMENTAL DESIGN

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GRANT FIT

Exploratory Aim 1 for conservation and causality; successful pilots support Aim 2 epistasis and Aim 3 spatial mechanism.

STRONGEST REVIEWER-FACING PREMISE

Xenopus can test when, where, and whether a newly reported cellular mechanism controls intact vertebrate development in one experimentally integrated system.

SUPPORTING SOURCES

featured paper: [Featured paper \(publisher-linked source\) 10.1126/science.aec0473](https://doi.org/10.1126/science.aec0473)

In search of past genetics

Stephen Chanock

journal-article | Science 393(6817) | 2026-09-17

DOI 10.1126/science.aek3832

Access: Publisher-deposited abstract retrieved through Crossref; no full-text review is claimed. | Evidence status: abstract-supported

MAIN DISCOVERY	Population data uncover the history of a rare genetic variant associated with high risk of lung cancer
WHY IT MATTERS	The study advances mechanistic understanding of In search of past genetics and creates a defined basis for testing conservation and developmental function in a vertebrate embryo.
MAIN / NEW METHODS	• Publisher-deposited title/abstract analysis; consult the linked paper for complete technical details
KEY EVIDENCE	• The retrieved publisher-deposited metadata identifies this item as a journal article in the detected volume and issue. • The DOI provides the canonical path for manual verification of figures and full methods.
LIMITATIONS	• This automated report used title, DOI metadata, and an available abstract; it does not claim manual full-text, figure, or complete methods review. • The deterministic extractor cannot assess experimental quality, effect size, statistics, or unreported negative results beyond the retrieved source text. • The proposed Xenopus mechanism is new speculation and has not been established by the featured paper.

Developmental timing and tissue autonomy of In search of past genetics

IMPORTANCE / SIGNIFICANCE

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SUPPORTING SOURCES

featured paper: [Featured paper \(publisher-linked source\) 10.1126/science.aek3832](#)

Multi-organelle signatures map cell-state diversity and metabolic adaptation in tissues

Raghabendra Adhikari, Alexander Hillsley, Alana Dowdell Johnson, Shihong Max Gao, Isabel Espinosa-Medina, Jan Funke, Daniel Feliciano

journal-article | Science 393(6817) | 2026-09-17

DOI 10.1126/science.ady6372

Access: Publisher-deposited abstract retrieved through Crossref; no full-text review is claimed. | Evidence status: abstract-supported

MAIN DISCOVERY	Cell-state diversity drives tissue adaptability, repair, and disease resilience, but capturing this complexity is a challenge.
WHY IT MATTERS	Intravital imaging linked fasting-induced organelle remodeling with altered mitochondrial membrane potential in vivo, supporting multi-organelle architecture as a structural readout of tissue adaptation.
MAIN / NEW METHODS	• Live-cell or intravital imaging • Quantitative imaging • Computational modeling
KEY EVIDENCE	• Cell-state diversity drives tissue adaptability, repair, and disease resilience, but capturing this complexity is a challenge. • In liver, sOrganellomics revealed that zonal position did not fully explain organelle-defined hepatocyte categories.
LIMITATIONS	• This automated report used title, DOI metadata, and an available abstract; it does not claim manual full-text, figure, or complete methods review. • The deterministic extractor cannot assess experimental quality, effect size, statistics, or unreported negative results beyond the retrieved source text. • The proposed Xenopus mechanism is new speculation and has not been established by the featured paper.

Developmental timing and tissue autonomy of Multi-organelle signatures map cell-state diversity and metabolic adaptation in...

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Disorder-promoted stability

Arthur N. Montanari, Pietro Zanin, Adilson E. Motter

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Access: Publisher-deposited abstract retrieved through Crossref; no full-text review is claimed. | Evidence status: abstract-supported

MAIN DISCOVERY	Here, we show that this conclusion arises from model reductions introduced for mathematical tractability and breaks down when nodal dynamics are higher-dimensional, yielding non-Hermitian Jacobians.
WHY IT MATTERS	Our framework reveals disorder not as a liability but as a general resource for stabilizing complex systems.
MAIN / NEW METHODS	• Publisher-deposited title/abstract analysis; consult the linked paper for complete technical details
KEY EVIDENCE	• Here, we show that this conclusion arises from model reductions introduced for mathematical tractability and breaks down when nodal dynamics are higher-dimensional, yielding non-Hermitian Jacobians. • Non-Hermiticity also underlies the stabilizing effects of network heterogeneity, which can arise even in one-dimensional nodal dynamics through nonreciprocal interactions, as shown for ecological networks. • Our framework reveals disorder not as a liability but as a general resource for stabilizing complex systems.
LIMITATIONS	• This automated report used title, DOI metadata, and an available abstract; it does not claim manual full-text, figure, or complete methods review. • The deterministic extractor cannot assess experimental quality, effect size, statistics, or unreported negative results beyond the retrieved source text. • The proposed Xenopus mechanism is new speculation and has not been established by the featured paper.

Developmental timing and tissue autonomy of Disorder-promoted stability

IMPORTANCE / SIGNIFICANCE

Gene Regulation And Chromatin is central to how embryonic cells convert molecular state into reproducible tissue organization. Establishing whether the featured mechanism operates during early vertebrate development could reveal a conserved control point and a tractable route to explain developmental variability without overstating clinical translation.

KNOWLEDGE GAP

The featured paper does not establish whether the focal regulator or mechanism described by the featured paper acts during early vertebrate development, whether its effect is cell-autonomous, which developmental window is sensitive, or whether molecular and morphogenetic phenotypes can be rescued independently.

RATIONALE AND XENOPUS ADVANTAGE

The paper supplies a candidate mechanism, while early *Xenopus* provides rapid external development, targeted blastomere injection, explants, lineage tracing, live imaging, inexpensive biological replication across clutches, and direct comparison of molecular and tissue-scale outcomes.

CENTRAL HYPOTHESIS

Speculative: the focal regulator or mechanism described by the featured paper acts cell-autonomously during stages 8-14 to stabilize pathway activity, lineage allocation, and spatial marker-domain size; acute loss will reduce the predicted molecular readout and increase pattern variance, while stage-matched wild-type rescue will restore both.

EXPERIMENTAL DESIGN

Use *Xenopus tropicalis* for F0 genetics and *Xenopus laevis* explants. Target animal-cap, organizer, and mesendodermal lineages by injecting two independent CRISPR RNPs at the 4- to 8-cell stage with a fluorescent lineage tracer; analyze stages 8-14. Groups: uninjected, non-targeting RNP, each guide alone, pooled guides, dose series, wild-type mRNA rescue, perturbation-resistant rescue, and temporally delayed perturbation where a validated small molecule or inducible construct exists. Primary readouts: pathway activity, lineage allocation, and spatial marker-domain size. Secondary readouts: cell behavior, developmental timing, and gastrulation morphology, amplicon sequencing, RT-qPCR or in situ hybridization, and an orthogonal biochemical or reporter assay. Use at least three independent spawnings, 20-30 embryos per group per spawning for morphology, and 8-10 pooled or individually profiled samples per molecular condition; randomize embryos, blind image scoring, and analyze clutch as a random effect with multiplicity-controlled planned contrasts. Decision point: advance to epistasis and spatial single-cell profiling only if both guides reproduce the effect and rescue restores at least 50% of the molecular and developmental phenotype.

EXPECTED RESULTS AND INTERPRETATIONS

Support would be a guide-concordant loss of the predicted molecular readout, increased spatial or embryo-to-embryo variance, and rescue of both molecular and tissue phenotypes. This would place the mechanism upstream of developmental robustness. A molecular phenotype without morphogenesis would suggest buffering or a parallel tissue effector and would redirect the next aim to sensitized or explant conditions. A morphogenetic phenotype without the predicted reporter change would argue for an alternative biochemical function. No reproducible phenotype after confirmed editing would refute an obligate early role and shift the model toward redundancy, later function, or species-specific deployment.

POTENTIAL PITFALLS

- F0 mosaicism may weaken or spatially blur phenotypes.
- Paralog redundancy or maternal stores may mask an early requirement.
- High-dose perturbation may cause nonspecific toxicity or developmental delay.

- The featured mechanism may not be conserved or may use different effectors in amphibians.
- Abstract-only evidence may omit critical context needed to choose the most specific assay.

ALTERNATIVE STRATEGIES

- Map editing per embryo and enrich analysis for high-editing specimens; establish stable alleles if mosaicism remains limiting.
- Target expressed paralogs jointly, use maternal-zygotic strategies where feasible, or test dominant-negative and degron-based acute perturbations.
- Use lower doses, later induction, tissue-restricted injection, explants, and matched survival/staging controls to separate mechanism from toxicity.
- Test *Xenopus laevis* and *X. tropicalis* orthologs, then validate the key interaction biochemically or in a complementary cell model.
- Before full-scale work, manually review the linked paper's figures and methods and revise the reporter or perturbation to match the demonstrated mechanism.

PUBLISHED ENABLING EVIDENCE AND PROPOSED PRELIMINARY-DATA PLAN

Published enabling evidence and proposed preliminary-data plan: no unpublished user-laboratory data are assumed. The featured paper is the cited cross-species enabling source; direct *Xenopus* evidence was not automatically verified. Pilot 1: map candidate expression or pathway activity at 5-6 early stages using RT-qPCR and in situ hybridization (three spawnings; expected Figure 1A-B); go if a reproducible spatial or temporal signal exceeds twofold over the lowest stage. Pilot 2: test two CRISPR RNPs plus non-targeting control in three spawnings with at least 25 embryos per group, amplicon sequencing, blinded morphology, and the primary molecular readout (Figure 1C-E); go if both guides produce the same directional effect of at least 20% with acceptable survival. Pilot 3: perform wild-type mRNA rescue in three spawnings with at least 20 embryos per group (Figure 1F-G); go if rescue reverses at least 50% of the molecular and phenotype effect. Failure at any threshold triggers the paired alternative strategy rather than a claim of preliminary support.

NOVELTY medium-high - transfers a recent mechanism into a spatially and temporally tractable vertebrate embryo	FEASIBILITY high for the pilots - injection, explants, imaging, genotyping, and rescue are routine <i>Xenopus</i> capabilities	RISK medium - conservation, maternal contribution, and mosaicism are explicit decision points
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GRANT FIT

Exploratory Aim 1 for conservation and causality; successful pilots support Aim 2 epistasis and Aim 3 spatial mechanism.

STRONGEST REVIEWER-FACING PREMISE

Xenopus can test when, where, and whether a newly reported cellular mechanism controls intact vertebrate development in one experimentally integrated system.

SUPPORTING SOURCES

featured paper: [Featured paper \(publisher-linked source\) 10.1126/science.aeg3946](#)

VIPR RNA-guided DNA recognition by noncontiguous geometric triplex formation

Peter H. Yoon, Trevor A. Docter, Zeyuan Terry Zhang, Kenneth Loi, Santiago C. Lopez, Luis E. Valentin-Alvarado, Owen T. Tuck, Stephen G. Brohawn, et al.

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MAIN DISCOVERY	These findings suggest that triplex-mediated target-strand handoff could enable noncontiguous and programmable RNA-guided DNA recognition in VIPR systems.
WHY IT MATTERS	These findings suggest that triplex-mediated target-strand handoff could enable noncontiguous and programmable RNA-guided DNA recognition in VIPR systems.
MAIN / NEW METHODS	• Electron microscopy • Quantitative imaging
KEY EVIDENCE	• These findings suggest that triplex-mediated target-strand handoff could enable noncontiguous and programmable RNA-guided DNA recognition in VIPR systems.
LIMITATIONS	• This automated report used title, DOI metadata, and an available abstract; it does not claim manual full-text, figure, or complete methods review. • The deterministic extractor cannot assess experimental quality, effect size, statistics, or unreported negative results beyond the retrieved source text. • The proposed Xenopus mechanism is new speculation and has not been established by the featured paper.

Developmental timing and tissue autonomy of VIPR RNA-guided DNA recognition by noncontiguous geometric triplex formation

IMPORTANCE / SIGNIFICANCE

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CROSS-PAPER SYNTHESIS

RECURRING MECHANISMS

- gene regulation and chromatin (8 papers)
- cell signaling and fate control (1 paper)
- immunity and host response (1 paper)
- metabolism and organelle function (1 paper)

METHODOLOGICAL TRENDS

- Publisher-deposited title/abstract analysis; consult the linked paper for complete technical details (5)
- Computational modeling (2)
- CRISPR genome editing (2)
- Quantitative imaging (2)
- Single-nucleus multi-omics (1)
- Genome-wide association or QTL mapping (1)

RANKED XENOPUS OPPORTUNITIES

#1 Developmental timing and tissue autonomy of Transcriptome-based classification in mice with ASD-risk mutations

Significance: high if the focal mechanism is conserved | Novelty: medium-high | Mechanistic Clarity: medium pending candidate-specific assay refinement | Preliminary Support: featured-paper evidence; Xenopus pilots proposed | Xenopus Advantage: high for temporal, spatial, rescue, and explant experiments | Feasibility: high for the three rapid pilots | Follow On Aims: epistasis, spatial profiling, and cross-species validation

#2 Developmental timing and tissue autonomy of Splenic regulation of systemic platelet activation state

Significance: high if the focal mechanism is conserved | Novelty: medium-high | Mechanistic Clarity: medium pending candidate-specific assay refinement | Preliminary Support: featured-paper evidence; Xenopus pilots proposed | Xenopus Advantage: high for temporal, spatial, rescue, and explant experiments | Feasibility: high for the three rapid pilots | Follow On Aims: epistasis, spatial profiling, and cross-species validation

#3 Developmental timing and tissue autonomy of A noncontiguous code for RNA-guided DNA recognition at the origin of CRISPR-Cas

Significance: high if the focal mechanism is conserved | Novelty: medium-high | Mechanistic Clarity: medium pending candidate-specific assay refinement | Preliminary Support: featured-paper evidence; Xenopus pilots proposed | Xenopus Advantage: high for temporal, spatial, rescue, and explant experiments | Feasibility: high for the three rapid pilots | Follow On Aims: epistasis, spatial profiling, and cross-species validation

#4 Developmental timing and tissue autonomy of A viral origin for RNA-guided immunity

Significance: high if the focal mechanism is conserved | Novelty: medium-high | Mechanistic Clarity: medium pending candidate-specific assay refinement | Preliminary Support: featured-paper evidence; Xenopus pilots proposed | Xenopus Advantage: high for temporal, spatial, rescue, and explant experiments | Feasibility: high for the three rapid pilots | Follow On Aims: epistasis, spatial profiling, and cross-species validation

COHERENT MULTI-AIM PROPOSAL

A coherent multi-aim proposal can unite papers centered on gene regulation and chromatin: Aim 1 establishes conservation and timing, Aim 2 resolves cell autonomy and pathway order, and Aim 3 links molecular activity to tissue-scale robustness.

INDEPENDENT PILOTS

- Developmental timing and tissue autonomy of Splenic regulation of systemic platelet activation state
- Developmental timing and tissue autonomy of A noncontiguous code for RNA-guided DNA recognition at the origin of CRISPR-Cas
- Developmental timing and tissue autonomy of A viral origin for RNA-guided immunity

SOURCE / METHOD NOTE

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